Solution of problems on motion under gravity:

1.

gap between two bodies = Displacement of 1st body in 3 sec − Displacement of 2nd body in 2 sec

$$= \frac{1}{2} g (3)^2 - \frac{1}{2} g (2)^2$$

$$= \frac{1}{2} \times 10 \times 9 - \frac{1}{2} \times 10 \times 4$$

$$= 45 - 20 = 25\,m.$$

2. Height of tower $h = ut + \frac{1}{2} g t^2$

$\because u = 0, \therefore h = \frac{1}{2} g t^2$

$$= \frac{1}{2} \times 10 \times (8)^2$$

$$= 320\,m.$$

$\frac{1}{4}$th of $320\,m = \frac{1}{4} \times 320 = 80\,m.$

so for 1st 80 m from top from rest

$$h = \frac{1}{2} g t^2$$

$$80 = \frac{1}{2} \times 10 \times t^2$$

$$t^2 = 16$$

$$t = 4\,sec$$

3. Distance covered in tth sec

$$h_t = u + \frac{1}{2} g (2t - 1)$$

$\because u = 0 \quad \therefore h_t = \frac{1}{2} g (2t - 1)$

$$\therefore \frac{h_{t_1}}{h_{t_2}} = \frac{2t_1 - 1}{2t_2 - 1}$$

$\because t_1 = 4$th sec, $t_2 = 5$th sec

$$\therefore \frac{h_4}{h_5} = \frac{2 \times 4 - 1}{2 \times 5 - 1} = \frac{7}{9}$$

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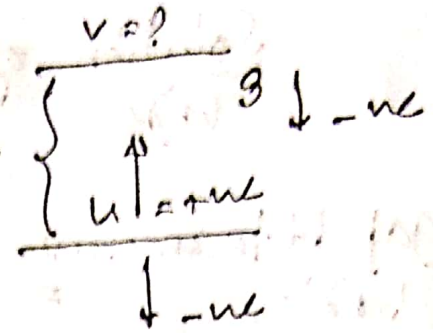
4.

$V = u + at$

$V = u - gt$

$V = 40 - 10 \times 2$

$\boxed{V = 20 m/s}$



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$v^2 = u^2 + 2as$

here $u = +10 m/s$

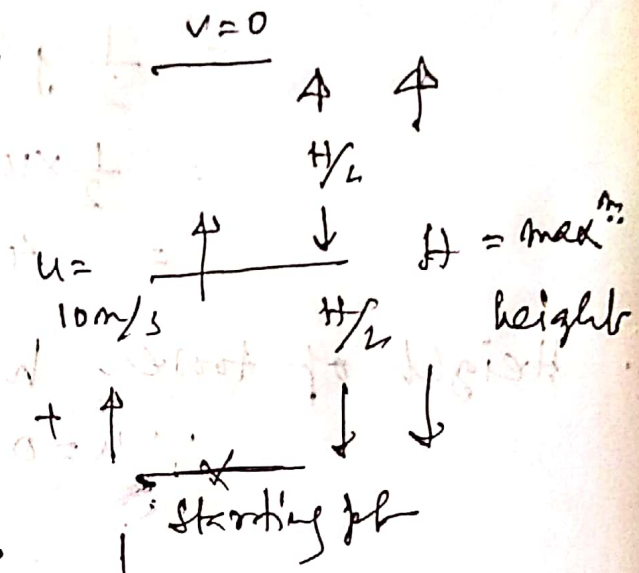
$v = 0$

$a = -g, \; s = H/2$

$0 = (10)^2 - 2g \, H/2$

$H = \frac{100}{g} = \frac{100}{10}$

$\boxed{H = 10 m}$



6/

$v^2 = u^2 + 2as$

$v = ?, \; u = 0, \; a = -g,$

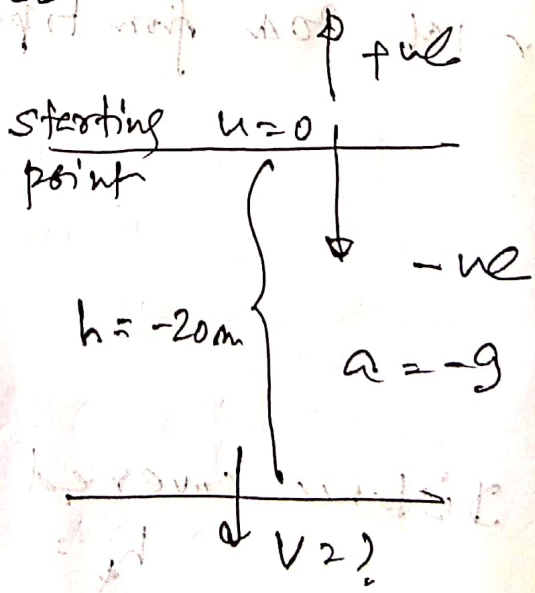
$s = -h = -20 m$

$v^2 = 2 \times (-g)(-h)$

$= 2 \times g \times h$

$v^2 = 2 \times 10 \times 20$

$\boxed{v = 20 m/s}$



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(7) Since they meet, so

Displacement of 1st ball in 18 sec = Displacement of 2nd ball in $(18-6) = 12$ sec

$\therefore \frac{1}{2} g (18)^2 = u \times 12 + \frac{1}{2} g \times (12)^2$

$\therefore u \times 12 = \frac{1}{2} g \{(18)^2 - (12)^2\}$

$\therefore u \times 12 = \frac{1}{2} \times 10 \times (18+12)(18-12)$

$\therefore u \times 12 = 5 \times 30 \times 6$

$\therefore u = \frac{5 \times \cancel{30}^{15} \times \cancel{6}}{\cancel{12}\ \cancel{2}}$

$\boxed{u = 75 \text{ m/s}}$

Note.

starting pt. u ↑ +ve

h = -ve (-ve) ↓ -ve

a = -g = -ve

so $s = ut + \frac{1}{2} g t^2$

becomes

$-h = -ut - \frac{1}{2} g t^2$

ie $\boxed{h = ut + \frac{1}{2} g t^2}$

for u = 0

$\boxed{h = \frac{1}{2} g t^2}$

(8) $h = ut + \frac{1}{2} g t^2$ $\because u = 0$, $\therefore \boxed{h = \frac{1}{2} g t^2}$

$\therefore t = \sqrt{\frac{2h}{g}}$

So time taken by A to fall thro' 16 m

$t_A = \sqrt{\frac{2 \times 16}{g}}$

& " " " B " " " 25 m

$t_B = \sqrt{\frac{2 \times 25}{g}}$

$\therefore \boxed{\frac{t_A}{t_B} = \sqrt{\frac{16}{25}} = \frac{4}{5}}$

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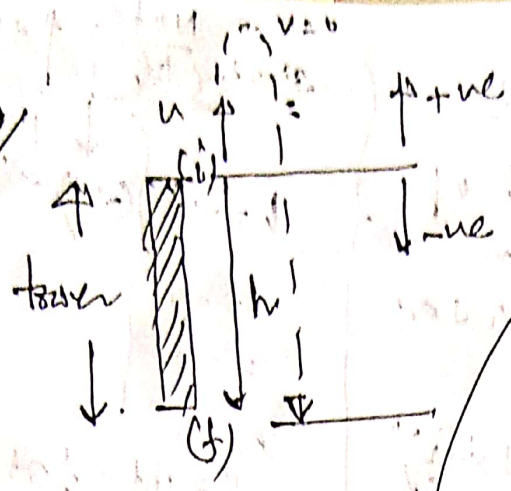


$S = ut + \frac{1}{2}at^2$

here, $S = h = -40m$,
$u = +10 m/s$
$a = -g = -10 m/s^2$
$t = ?$

$\therefore -40 = 10t - \frac{1}{2} \times 10 t^2$

$5t^2 - 10t - 40 = 0$

$t^2 - 2t - 8 = 0$

$(t-4)(t+2) = 0$

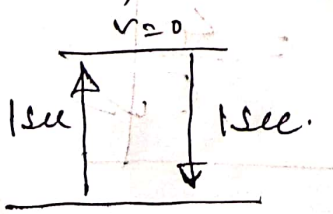
$\therefore t = 4 sec$ or $t = -2 sec$. (not possible)

$\therefore \boxed{t = 4 sec}$

10/ Next ball is thrown upward when the velocity of the ball thrown earlier is zero, ie & balls are thrown at regular interval of 1 sec.

It means time taken by each ball to reach max^m height = 1 sec.

Now, Since time of ascent
(সর্বোচ্চ বিন্দুতে উঠতে সময়)



= time of descent
(সর্বোচ্চ বিন্দু থেকে নামতে সময়)

$\therefore$ time of descent from top = 1 sec.

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Now, during downward journey from topmost position $u = 0$

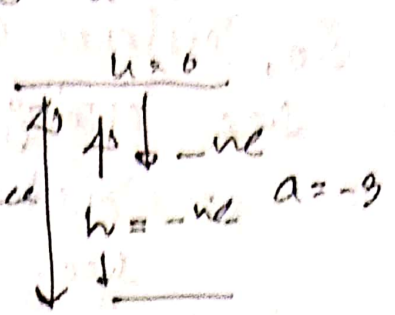
$$s = \frac{1}{2}at^2$$

$$-h = \frac{1}{2}(-g)t^2$$

$$h = \frac{1}{2}gt^2$$

$$= \frac{1}{2} \times 10 \times 1^2$$

$$= 5m$$



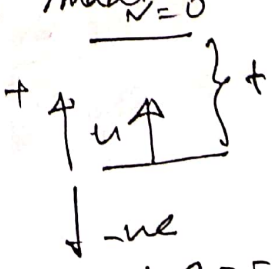
Note max. height এ গেলে বস্তু থেমে যায়, সেজন্য এই ধরনের problem solve করার সময়, যখন topmost point এসে থামে তখন $u = 0$ নিয়ে চলতি, কারণ সুবিধার জন্য $u = 0$ এবং $h = \frac{1}{2}gt^2$ এই সূত্র প্রযোজ্য হবে। কিন্তু যখন initial velocity problem এ দেওয়া থাকে, তখন সেই সূত্র দিয়ে কষে নিতে হবে।

example problem on this note concept

Exp. (12) A body is thrown vertically upward with initial velocity 30 m/s. Find distance covered by the body in last one second.

Method 1 এর মতো করে প্রথম problem এর মতো, এর এই Method নেই,

Let time taken to reach maximum height be t sec.



$$\therefore v = u + at$$

$$0 = u - gt$$

$$t = u/g = 30/10 = 3 \text{ sec.}$$

$\therefore a = -g$

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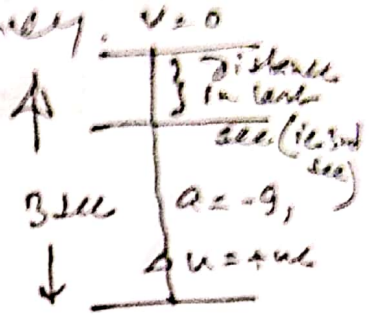
It means time of ascent is 3 sec

So, distance covered in last one sec during upward journey

= Distance covered in 3rd sec in upward journey.



} Distance in last sec (ie 3rd sec)

$$h_t = u - \frac{1}{2} g (2t-1)$$

$$h_3 = 30 - \frac{1}{2} \times 10 (2 \times 3 - 1)$$

$$h_3 = 30 - 5 \times 5 = 5m$$

Alternative Method

এর concept হচ্ছে এটাই, এর 1 step এ নামব হচ্ছে →

Distance travelled in last sec in upward journey

= Distance travelled in first one second during downward journey

Self this method

$$= \frac{1}{2} g t^2 \quad [\because \text{during downward journey from top or max}^m \text{ ht } u=0]$$

$$= \frac{1}{2} \times 10 \times (1)^2$$

$$= 5m$$

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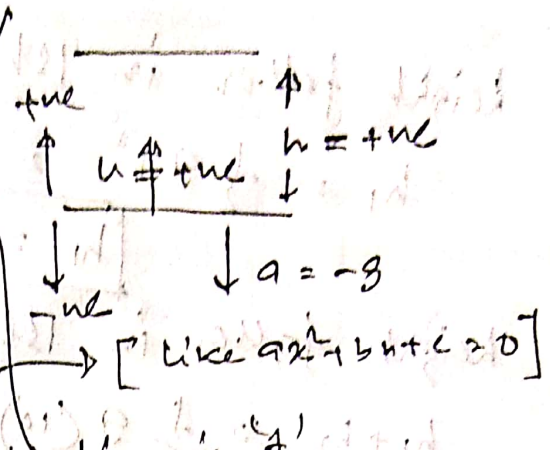
(11) $S = ut + \frac{1}{2}at^2$

$h = ut - \frac{1}{2}gt^2$

or

$$\boxed{\underbrace{g}_{a}t^2 - \underbrace{2u}_{b}t + \underbrace{2h}_{c} = 0}$$



[like $ax^2 + bx + c = 0$]

It is a quadratic equation in 't'.
ie t has two roots say t_1 & t_2
[it means particle comes to same height 'h' in two different times t_1 & t_2]

Note
$ax^2 + bx + c = 0$
— a quadratic eqn in 'x', If x has two roots 'α' & 'β' then

(*) $\alpha + \beta = -\frac{b}{a}$

(*) $\alpha\beta = \frac{c}{a}$

(*) $\therefore t_1 + t_2 = -\frac{(-2u)}{g}$

$t_1 + t_2 = \frac{2u}{g}$

$\therefore \boxed{u = \frac{g}{2}(t_1 + t_2)}$

& (*) $t_1 t_2 = \frac{2h}{g}$

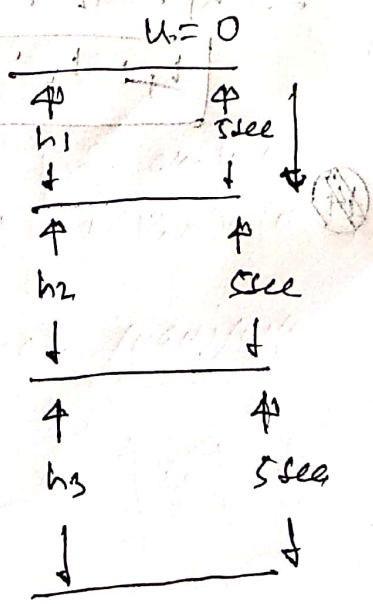
$\therefore \boxed{h = \frac{1}{2}g t_1 t_2}$

(13) If we consider downward direction -ve, then $h = -ve$, $a = -g$
& if $u = 0$

then $S = \frac{1}{2}at^2$

becomes $-h = \frac{1}{2}(-g)t^2$

$\boxed{h = \frac{1}{2}gt^2}$



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height fallen in 1st 5 sec

$h_1 = \frac{1}{2} g t^2 = \frac{1}{2} \times 10 \times 25$

$\boxed{h_1 = \cancel{75\,m.}\ 125\,m}$

height fallen in 1st 10 sec

$h_1 + h_2 = \frac{1}{2} g (10)^2 = 5 \times 100$

$= 500\,m$

$\therefore h_2 = (h_1 + h_2) - h_1 = 500 - 125$

$\boxed{h_2 = 375\,m}$

height fallen in 1st 15 sec

$h_1 + h_2 + h_3 = \frac{1}{2} g (15)^2 = 5 \times 225$

$= 1125\,m.$

$\therefore h_3 = (h_1 + h_2 + h_3) - (h_1 + h_2)$

$= 1125 - 500$

$\boxed{h_3 = 625\,m}$

$\therefore h_1 : h_2 : h_3 = 125 : 375 : 625$

$\boxed{h_1 : h_2 : h_3 = 1 : 3 : 5}$

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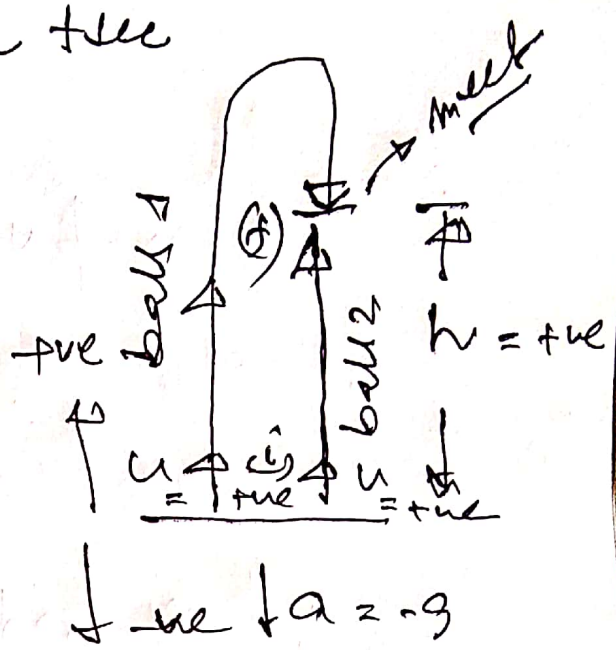
(14) Let two ball meets after time (after 1st ball is thrown).

Since two balls meet,

So

Displacement of ball 1 = Displacement of ball 2



$$ut - \frac{1}{2}gt^2 = u(t-T) - \frac{1}{2}g(t-T)^2$$

$$\cancel{ut} - \cancel{\frac{1}{2}gt^2} = \cancel{ut} - uT - \cancel{\frac{1}{2}gt^2} + gtT - \frac{1}{2}gT^2$$

$$gtT = uT + \frac{1}{2}gT^2$$

$$gt = u + \frac{gT}{2} \quad (\because T \neq 0)$$

$$\boxed{t = \frac{u}{g} + \frac{T}{2}}$$

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